

# Consumer Lens — Technical Architecture & Deployment Documentation

Smart India Hackathon 2024 · Problem Statement ID: 26034  
Automated Compliance Verification of Packaged Commodities under Legal Metrology Rules, 2011

Ministry of Consumer Affairs, Food & Public Distribution

Department of Consumer Affairs (DoCA) · Legal Metrology Division

Problem Statement ID: 26034 · Automated Compliance Verification of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011

## 1. Executive Summary & Objective

**Consumer Lens** is an institutional-grade, AI-powered regulatory enforcement platform engineered for Legal Metrology officers, field inspectors, and supervisory authorities across India.

The platform automates the inspection of physical packaged commodities and e-commerce product listings to ensure strict adherence to the **Legal Metrology Act, 2009** and the **Legal Metrology (Packaged Commodities) Rules, 2011 (LMPC Rules, 2011)**.

### Key Capabilities

- Multi-Modal Label Evidence Capture:** Live camera ingestion, high-resolution packaging photo uploads, and automated PDP web scraping for major e-commerce portals (Amazon, Flipkart).
- Vision AI Multimodal Extraction:** Gemini Vision multimodal inference detecting normalized 2D bounding boxes (`box_2d: [ymin, xmin, ymax, xmax]`), font size compliance, and print readability.
- Rule 7 & Table I Font Size Compliance:** Evaluates numeral and letter heights on the Principal Display Panel (PDP) against Table I statutory minimums, aspect ratio limits, and prominence of MRP.
- Rule 9 Manner of Declaration & Readability:** Real-time evaluation of print contrast adequacy, surface glare, blur, and statutory language compliance (Hindi/English).
- Rule 18(2) & Section 18 Fair Trade & Deceptive Declarations:** Automatic detection of unauthorized price sticker overlays, price alterations, smudging, and contradictory claims.
- Statutory Enforcement Dossier & PDF Memorandum:** Generates courtroom-ready digital inspection dossiers compliant with Section 65B of the Indian Evidence Act, complete with SHA-256 evidence hashes, verification QR codes, and institutional seals.
- Zero-Data-Loss Offline Resilience:** Client-side IndexedDB persistence ensures field inspectors never lose in-progress scans upon browser reloads or network dropouts.

## 2. High-Level System Architecture

The following diagram illustrates the end-to-end architecture of Consumer Lens, connecting the field inspector interface with the application server, AI vision inference engine, relational persistence layer, and external verification infrastructure.

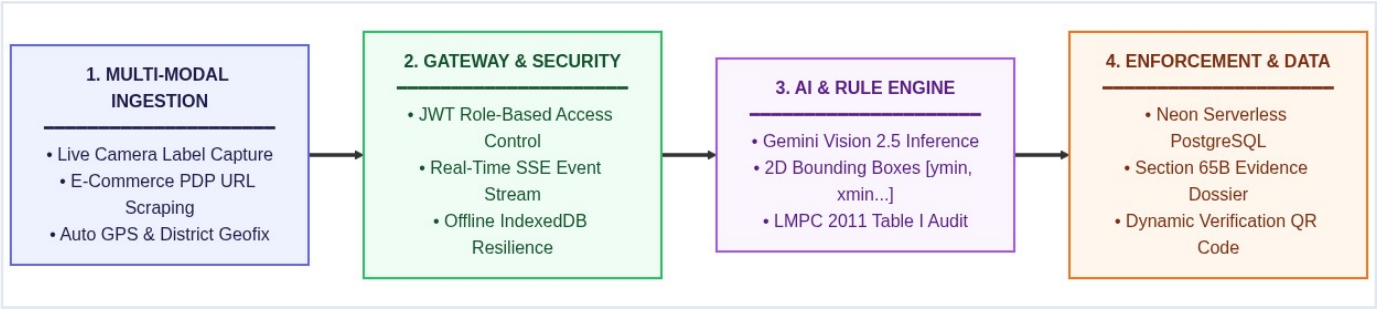


Figure 1: End-to-End System Architecture (4-Tier Ingestion to Enforcement)

### 3. System Design & User Interaction Flow

The following System Design cum User Flow diagram illustrates the exact synchronization between field inspector actions on the client interface and the asynchronous cloud services executing beneath the surface.

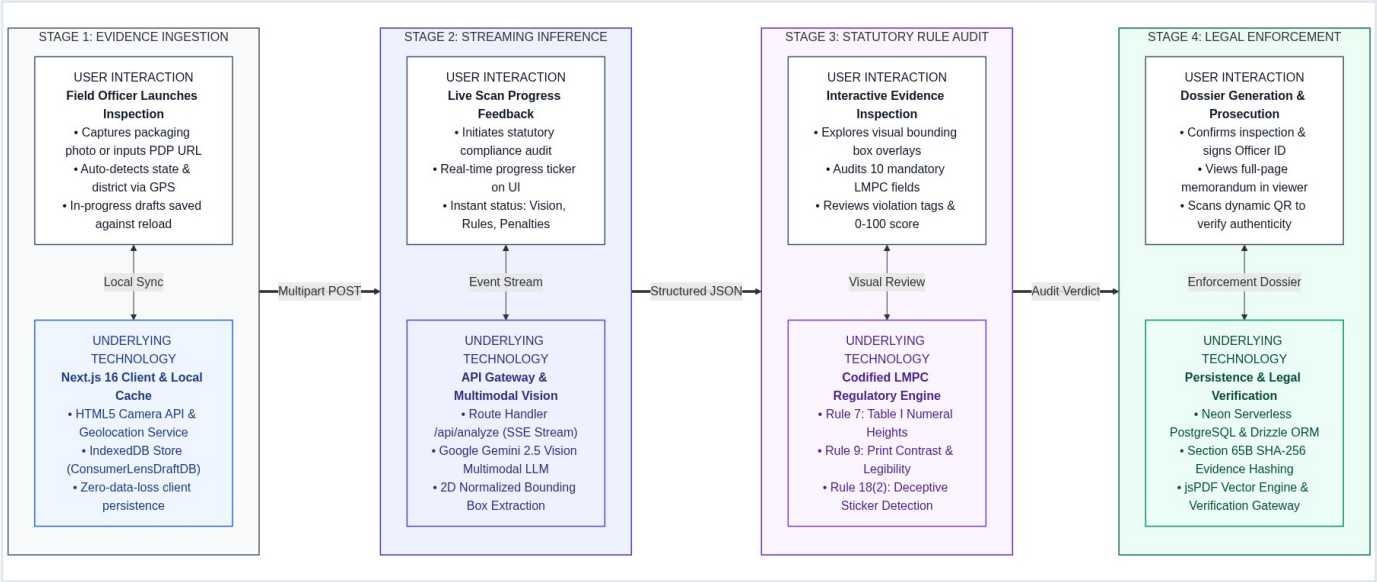


Figure 2: System Design cum User Interaction Flow (Field Officer vs. Cloud Services)

#### Stage-by-Stage Technical Interaction Matrix

| Stage                  | Field Inspector Interaction                                                                                                             | Technology Components Utilized                                                                                    | Data Processing & Protocol                                                                                                                   |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Evidence Ingestion  | Launches PWA on mobile or workstation; triggers live camera stream or inputs e-commerce PDP URL; state/district detected automatically. | HTML5 MediaDevices API, BigDataCloud Geocoding, client-side IndexedDB (ConsumerLensDraftDB).                      | Image serialized into Base64; draft state preserved locally to prevent loss on browser reload.                                               |
| 2. Streaming Inference | Clicks "Run Compliance Analysis"; views live status ticker advancing through inference milestones.                                      | Next.js Edge / Node Route Handlers (/api/analyze), Server-Sent Events (text/event-stream), Gemini Vision 2.5 API. | HTTP POST with multipart payload; server dispatches prompt + image to Gemini; streams back real-time step events.                            |
| 3. Statutory Audit     | Navigates visual label inspector; toggles 2D bounding boxes over packaging; audits missing declarations and penalty scores.             | React 19 Canvas/SVG coordinate mapper, LMPC Rule 7 Table I height engine, Rule 9 contrast scorer.                 | Maps normalized coordinates [ymin, xmin, ymax, xmax] to screen pixels; evaluates declarations against statutory rules; computes 0-100 score. |

|                      |                                                                                                                           |                                                                                           |                                                                                                                                  |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 4. Legal Enforcement | Confirms findings; saves inspection to state repository; launches full-screen PDF dossier; verifies tamper-proof QR code. | Neon Serverless PostgreSQL, Drizzle ORM, Web Crypto SHA-256 hashing, jsPDF vector engine. | Generates cryptographic hash over inspection parameters; stores record in Postgres; issues Section 65B certified PDF memorandum. |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|

## 4. LMPC Rules, 2011 Regulatory Decision Engine

The regulatory engine evaluates 10 statutory declaration fields against codified legal requirements from the Legal Metrology (Packaged Commodities) Rules, 2011:

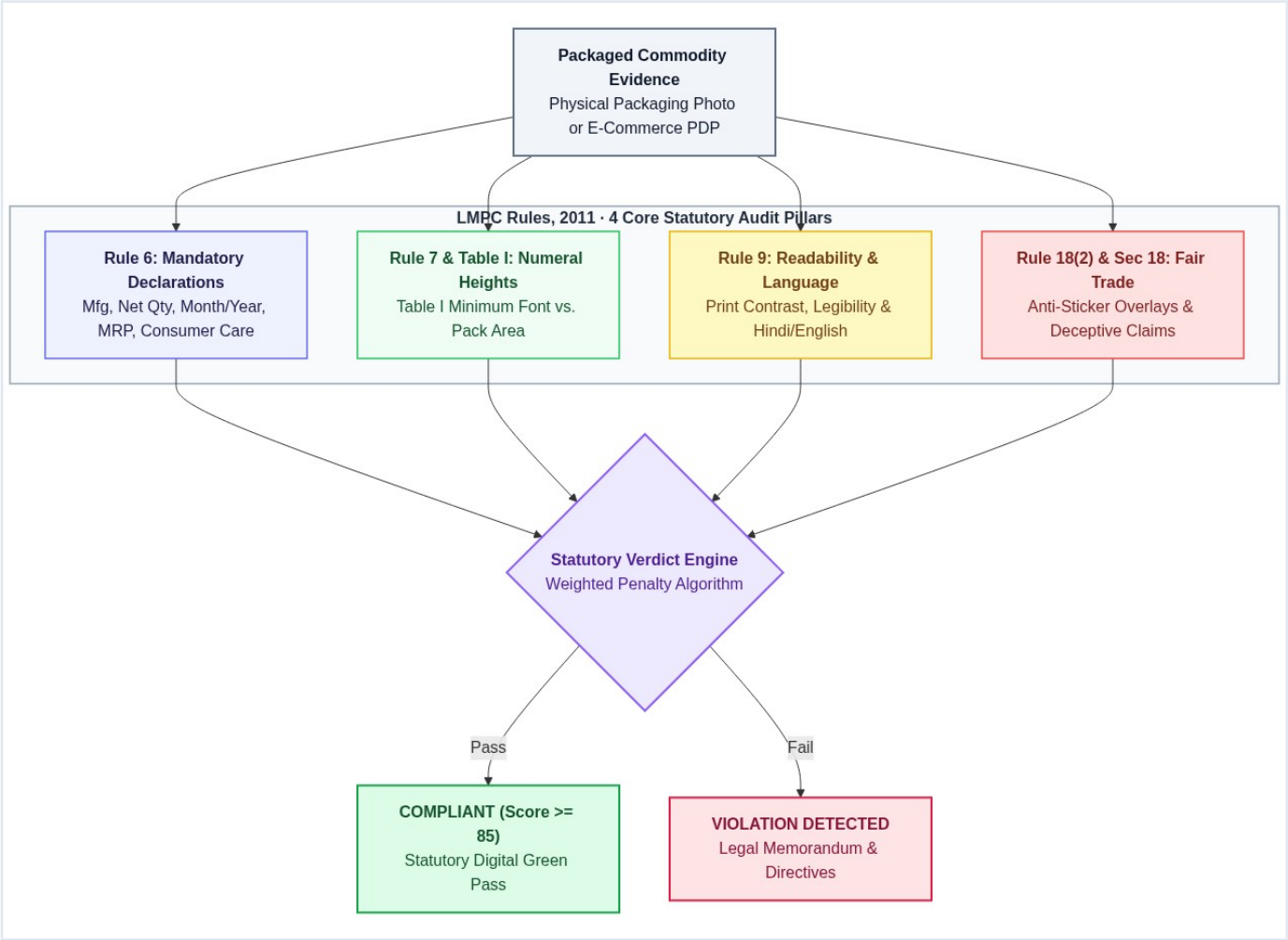


Figure 3: Codified LMPC Rules, 2011 Regulatory Decision Engine

### Statutory Severity Matrix

| LMPC Rule Citation | Mandatory Declaration            | Defect Condition                                                 | Severity Rating | Penalty Deduction |
|--------------------|----------------------------------|------------------------------------------------------------------|-----------------|-------------------|
| Rule 6(1)(a)       | Manufacturer / Packer / Importer | Missing name, missing registered address, or no role qualifier   | CRITICAL        | -25 pts           |
| Rule 6(1)(b)       | Generic / Common Name            | Brand name only; generic commodity nature omitted                | MAJOR           | -15 pts           |
| Rule 6(1)(c)       | Net Quantity                     | Non-SI units, missing numeral/unit, or missing space around unit | CRITICAL        | -25 pts           |
| Rule 6(1)(d)       | Date of Manufacture / Packing    | Missing month and year of packaging                              | MAJOR           | -15 pts           |

|                     |                                       |                                                                     |          |         |
|---------------------|---------------------------------------|---------------------------------------------------------------------|----------|---------|
| Rule 6(1)(da)       | Best Before / Use By Date             | Missing on perishable products, or missing qualifier                | MAJOR    | -15 pts |
| Rule 6(1)(e)        | Maximum Retail Price (MRP)            | Missing "Inclusive of all taxes", missing currency symbol           | CRITICAL | -25 pts |
| Rule 6(1)(f) / 6(2) | Consumer Care Grievance Cell          | Missing official phone, email, or physical address                  | MAJOR    | -15 pts |
| Rule 6(1)(g)        | Country of Origin                     | Missing country of origin on imported packaged goods                | MAJOR    | -15 pts |
| Rule 6(11)          | Unit Sale Price (USP)                 | Missing calculated unit price per gram/ml for packs > 100g/ml       | MINOR    | -10 pts |
| Rule 7 & Table I    | Minimum Font Size (PDP Height)        | Numeral/letter height below statutory Table I minimum for pack area | MAJOR    | -15 pts |
| Rule 9              | Print Legibility, Contrast & Language | Low print contrast, glare, blur, or non-English/Hindi declaration   | MINOR    | -10 pts |
| Rule 18(2) / Sec 18 | Misleading & Altered Pricing          | Dual pricing stickers, smudged MRP, deceptive claims                | CRITICAL | -30 pts |

## 5. Security & Role-Based Access Control (RBAC) Architecture

Consumer Lens implements institutional role segregation aligned with statutory enforcement hierarchies:

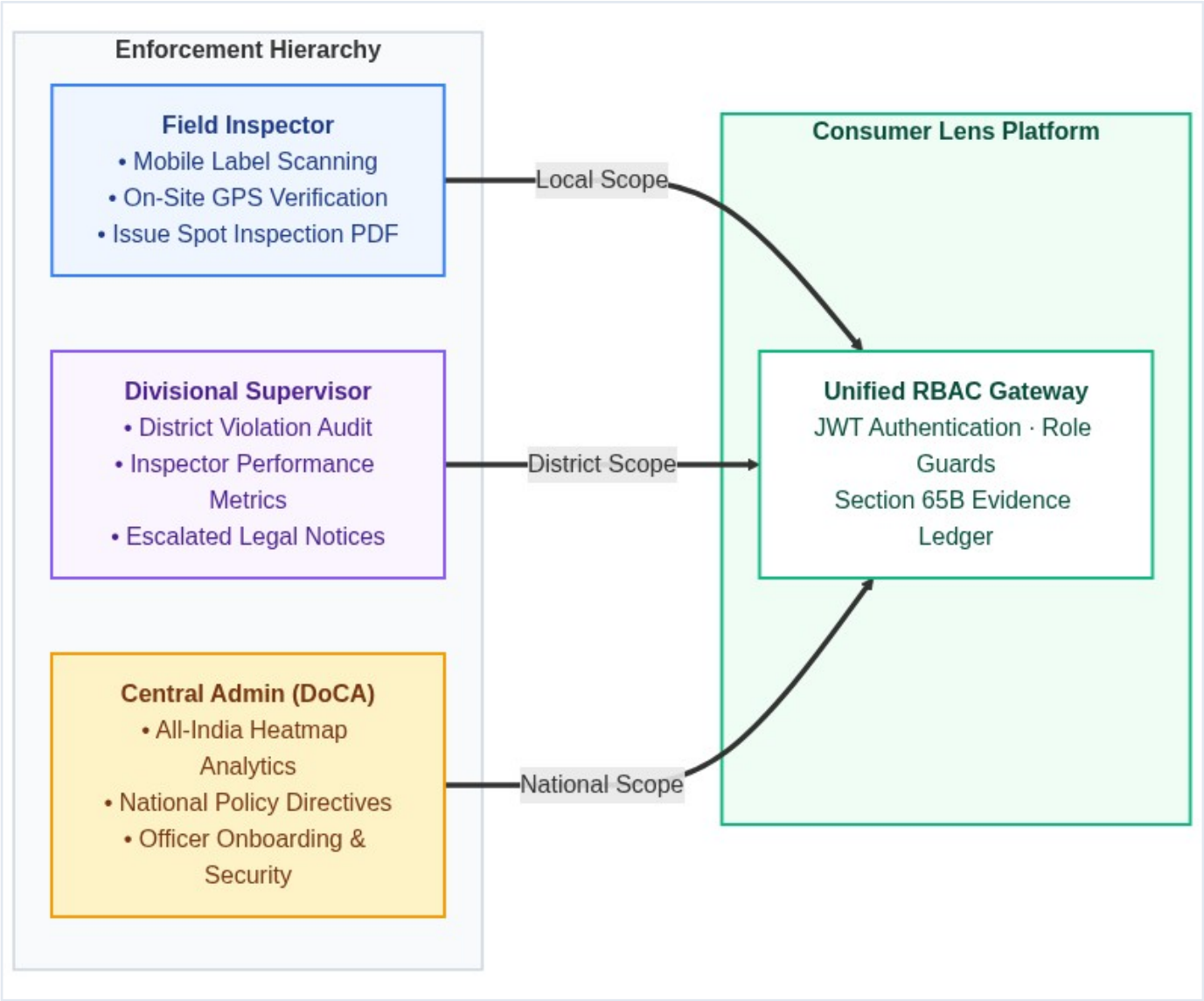


Figure 4: Role-Based Access Control (RBAC) & Enforcement Hierarchy

## 6. Section 65B Indian Evidence Act Compliance Framework

To ensure that generated inspection reports are legally admissible as electronic evidence in Indian courts of law during prosecution of non-compliant manufacturers, the platform incorporates a statutory digital evidence chain:

### 1. SHA-256 Cryptographic Evidence Digest:

```
\text{Digest} = \text{SHA-256}(\text{InspectionID} \parallel \text{Date} \parallel \text{Manufacturer} \parallel \text{MRP} \parallel \text{ImageBase64})
```

This cryptographic hash is stamped into the official dossier reference header. Any tampering with the document or evidence image invalidates the hash.

### 2. Dynamic Verification QR Codes:

Each generated memorandum contains vector QR codes pointing directly to [https://consumer-lens.vercel.app/inspections/\[id\]](https://consumer-lens.vercel.app/inspections/[id]). Enforcement officers, magistrates, and manufacturers can scan the QR code to verify the report's authenticity against the tamper-proof server record.

### 3. Official Seals & Digital Signatures:

The report renders proportional vector seals:

- National Emblem of India (State Emblem)
- 75 Azadi Ka Amrit Mahotsav Logo
- Food Safety and Standards Authority of India (FSSAI) Seal
- Legal Metrology Directorate Official Inspection Stamp
- Cryptographic vector checkmark and officer employee ID stamp.

## 7. Production Deployment & Cloud Framework

Consumer Lens is engineered for zero-downtime multi-cloud deployment with containerized support for Google Cloud Run, AWS ECS, or Vercel Edge.

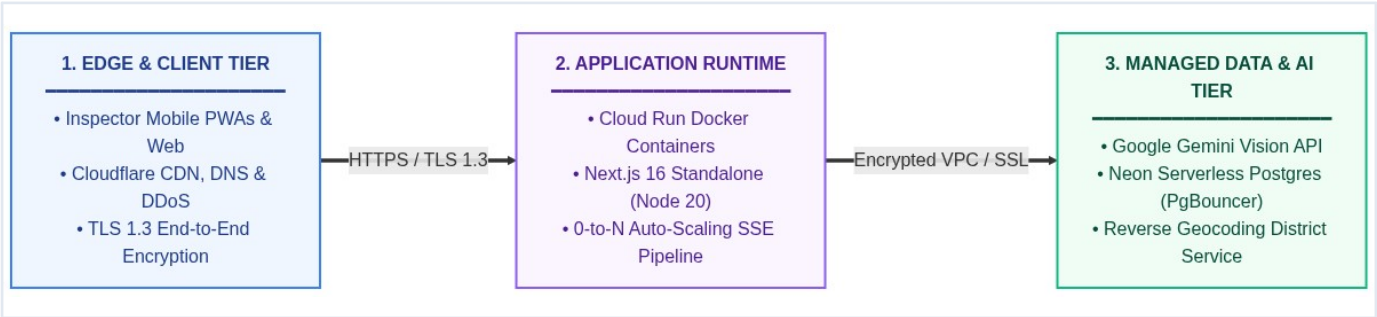


Figure 5: Multi-Cloud High-Availability Deployment Architecture

### Multi-Stage Dockerfile Architecture

The application includes a production-hardened multi-stage Docker build optimized for minimal image footprint and Google Cloud Run execution:

```
# Stage 1: Base Image with Node 20 Alpine & Corepack PNPM
FROM node:20-alpine AS base
WORKDIR /app
RUN corepack enable && corepack prepare pnpm@latest --activate

# Stage 2: Dependencies Installation with Cache Mounting
FROM base AS deps
RUN apk add --no-cache libc6-compat
WORKDIR /app
COPY package.json pnpm-lock.yaml* ./
RUN pnpm i --frozen-lockfile

# Stage 3: Builder with Next.js Standalone Optimization
FROM base AS builder
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY . .
ENV NEXT_TELEMETRY_DISABLED=1
ENV NODE_ENV=production
RUN pnpm build

# Stage 4: Minimal Runner (Distroless-style Non-Root Execution)
FROM node:20-alpine AS runner
WORKDIR /app
ENV NODE_ENV=production
ENV PORT=8080
ENV HOSTNAME="0.0.0.0"
RUN addgroup --system --gid 1001 nodejs && adduser --system --uid 1001 nextjs
COPY --from=builder /app/public ./public
COPY --from=builder --chown=nextjs:nodejs /app/.next/standalone ./
COPY --from=builder --chown=nextjs:nodejs /app/.next/static ./next/static
USER nextjs
EXPOSE 8080
CMD ["node", "server.js"]
```

## 8. Relational Schema & Entity-Relationship Model

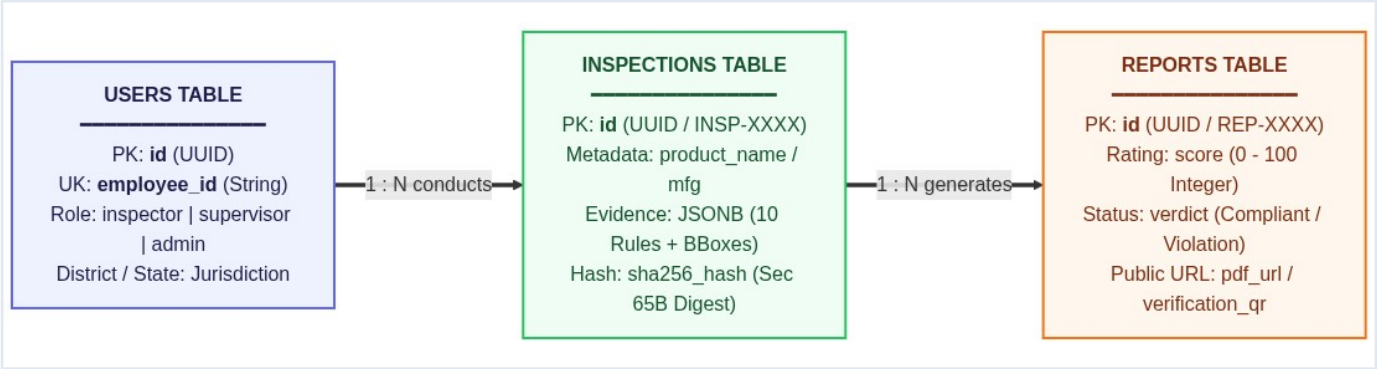


Figure 6: Relational Schema & Entity-Relationship Data Model

## 9. Presentation Cheat-Sheet (Summary for Evaluation & Slides)

| Metric / Dimension  | Implementation Detail                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Statutory Law       | Legal Metrology Act, 2009 & Legal Metrology (Packaged Commodities) Rules, 2011                            |
| Target Authority    | Department of Consumer Affairs (DoCA), Ministry of Consumer Affairs, Govt. of India                       |
| Front-End Stack     | Next.js 16.3 (App Router), React 19, Tailwind CSS 4, shadcn/ui, Lucide Icons                              |
| AI Inference        | Gemini Vision 2.5 Multimodal LLM returning normalized 2D bounding boxes ( <a href="#">box_2d</a> )        |
| Font Size Rules     | Rule 7 & Table I (Minimum numeral and letter height verification against packaging area)                  |
| Readability Rules   | Rule 9 (Automated contrast ratio, prominence, Hindi/English language, and print quality scoring)          |
| Deceptive Claims    | Rule 18(2) & Section 18 (Price alteration, sticker overlay detection, dual pricing & deceptive packaging) |
| Offline Resilience  | Native browser IndexedDB ( <a href="#">ConsumerLensDraftDB</a> ) caching images & scan state              |
| Legal Admissibility | Section 65B Indian Evidence Act SHA-256 evidence digest & dynamic verification QR                         |
| Database            | PostgreSQL with Drizzle ORM, connection pooling via Neon serverless                                       |
| Security            | JWT HTTP-only cookies, bcrypt hashing, role-based route middleware (Inspector/Supervisor/Admin)           |
| Containerization    | Multi-stage Dockerfile running as non-root <a href="#">nextjs</a> user on Alpine Node 20 (Port 8080)      |